

PUBLISHED: 03-MAY-2017
2014.0 RANGE ROVER (LG), 303-01

ENGINE - V8 S/C 5.0L PETROL

FUEL PUMP CAMSHAFT TIMING CHECK (G1473761)

GENERAL PROCEDURES

19.90.16	HIGH PRESSURE FUEL TIMING CHECK	5000 CC, AJ V8, SUPERCHARGED	1	USED WITHINS	
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SPECIAL TOOL(S)



PART(S)

STEP	PART NAME	QUANTITY
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Check Step 4	Oil pan drain plug seal	1
Check Step 8	Brake vacuum pump O-ring seal	1

CHECK

1. Disconnect the battery ground cable.
Refer to: Specifications (414-00 Battery and Charging System - General Information, Specifications).

2.  **WARNING:**

Make sure to support the vehicle with axle stands.

Raise and support the vehicle.

3. Refer to: Engine Undershield (501-02 Front End Body Panels, Removal and Installation).

4.  **CAUTION:**

Allow at least 10 minutes for the engine oil to drain.

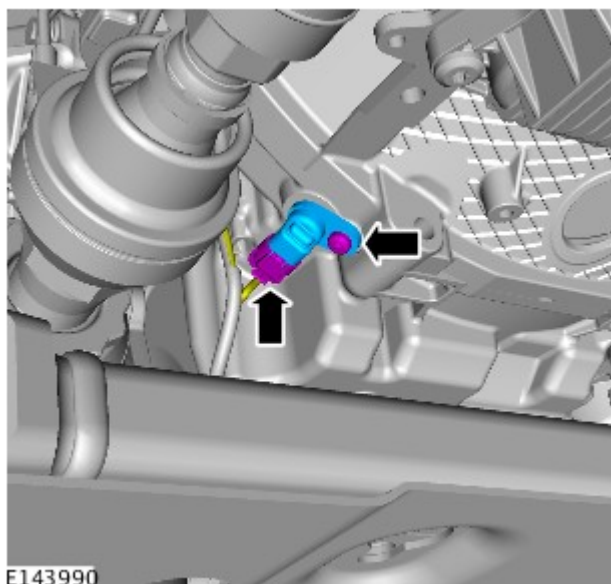
NOTES:

- Discard the sealing washer.
- Collect the engine oil in a clean container.



Renew Part: *Oil pan drain plug seal* Quantity: 1 .

5.



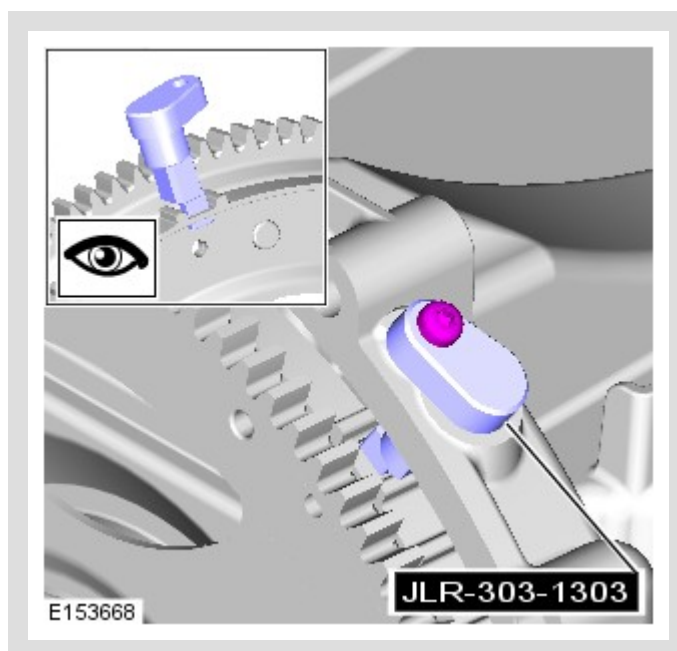
6.



CAUTIONS:

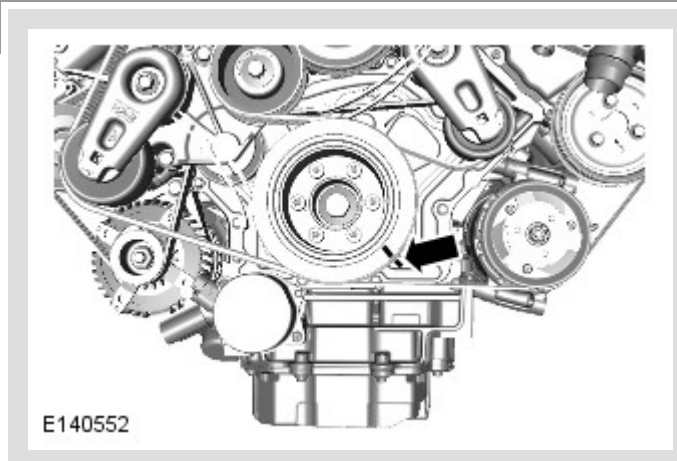
- Only rotate the crankshaft clockwise.

- Make sure that the crankshaft is fully locked.



- *Special Tool(s):* [JLR-303-1303](#)

7.



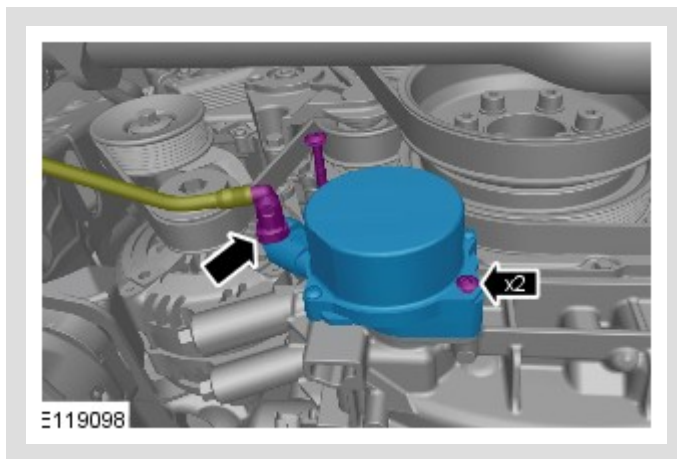
Using a suitable marker, mark the position of the crankshaft pulley as illustrated.

8.



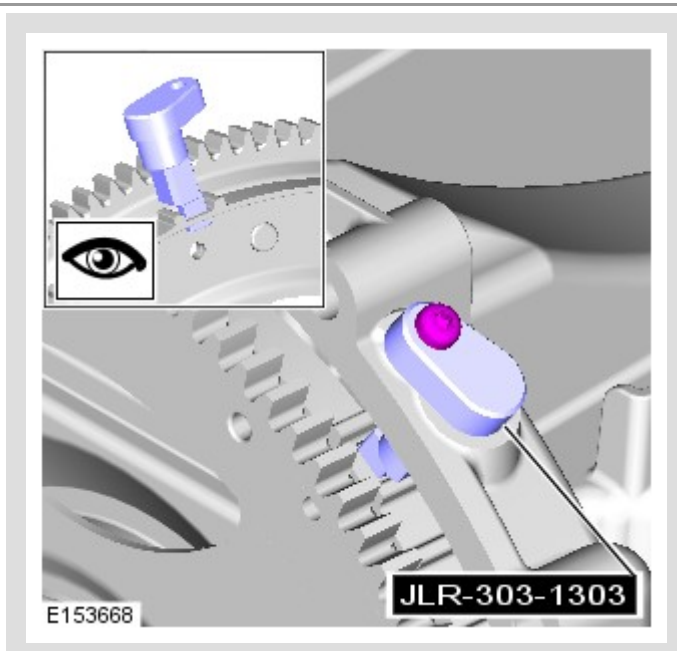
CAUTION:

Discard the seal.



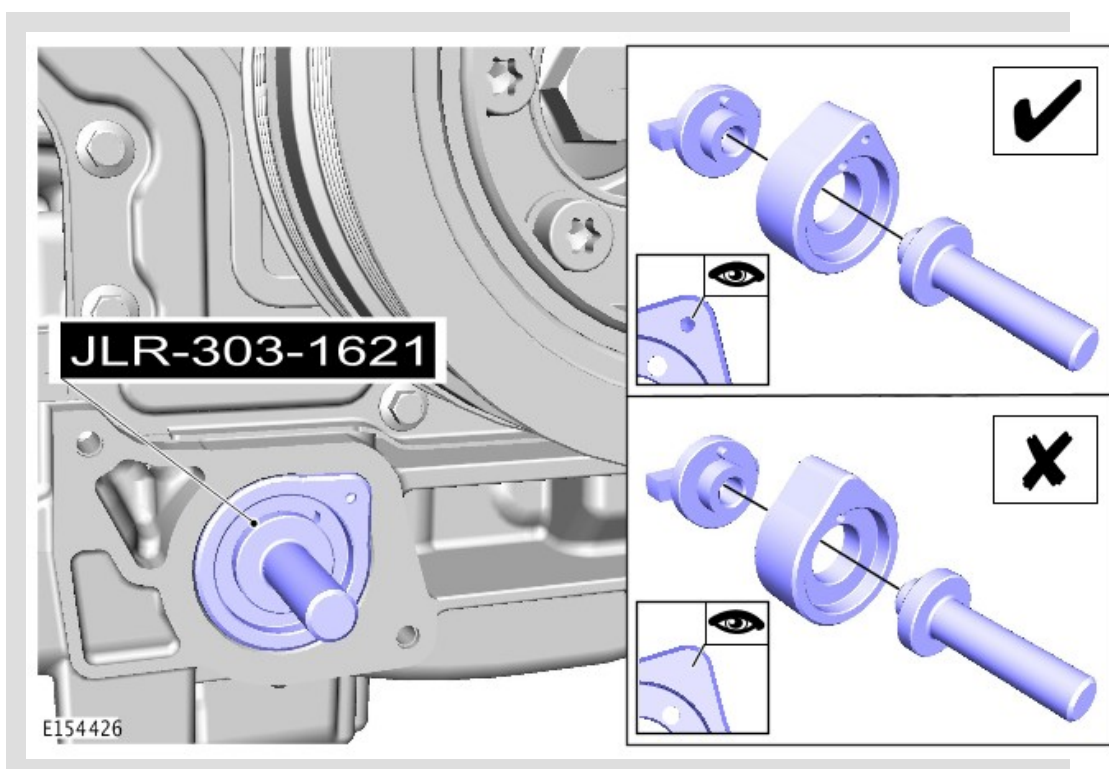
Renew Part: *Brake vacuum pump O-ring seal* Quantity: *1* .

9.



- Remove the special tool.

10.



If required, carefully adjust the crankshaft position to allow correct installation of the special tool.

Special Tool(s): [JLR-303-1621](#)

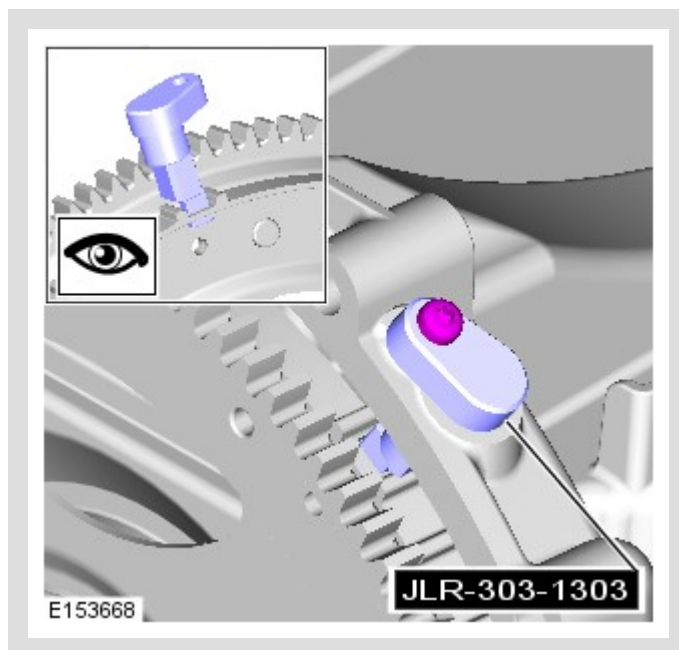
11.

⚠ CAUTION:

Do not use excessive force when adjusting the crankshaft position.

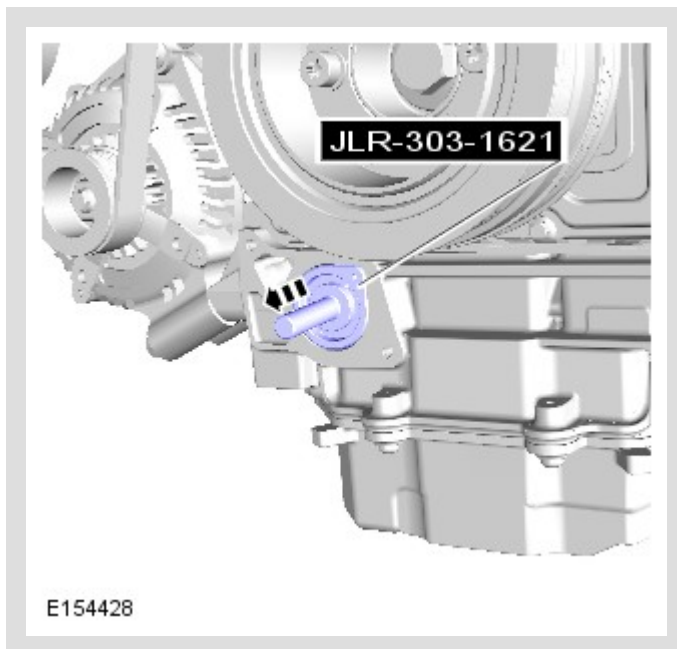
△ NOTE:

If the crankshaft timing tool cannot be installed, adjustment of the fuel pump camshaft timing will be required.



- If required, carefully adjust the crankshaft position to allow correct installation of the special tool.

Special Tool(s): [JLR-303-1303](#)

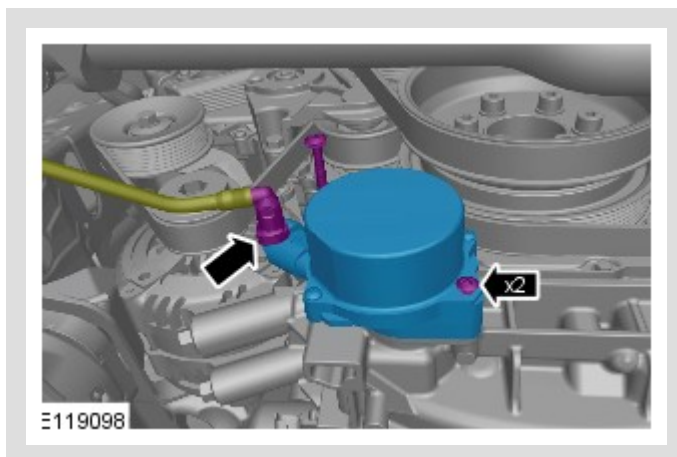


Special Tool(s): [JLR-303-1621](#)

13.

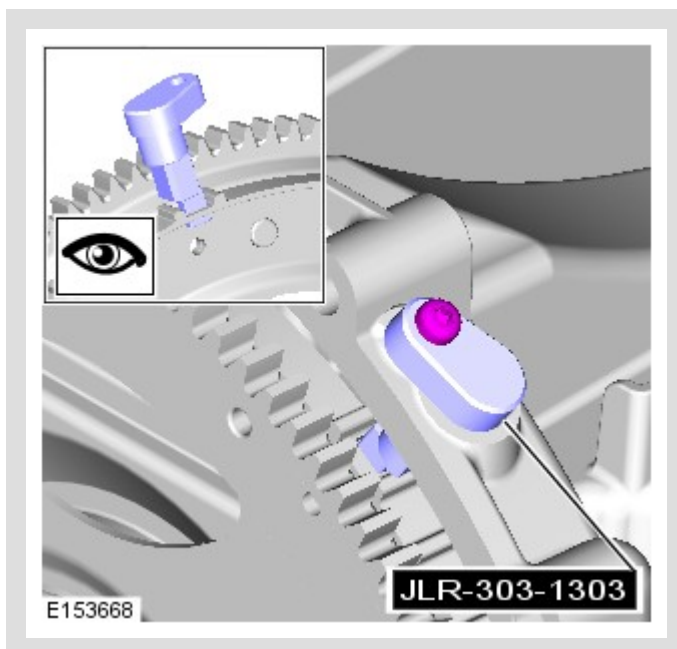
⚠ CAUTION:

Install a new seal.



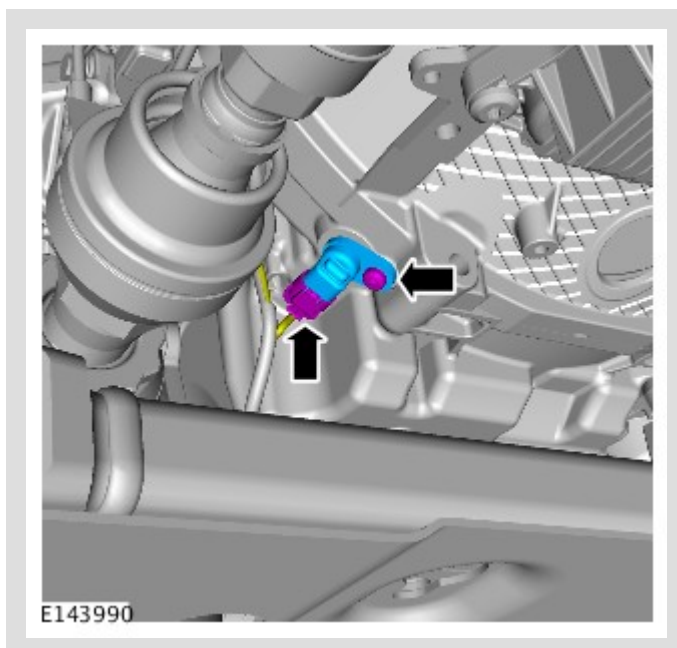
- Torque: 12 Nm

14.



- Remove the special tool.
Special Tool(s): [JLR-303-1303](#)

15.

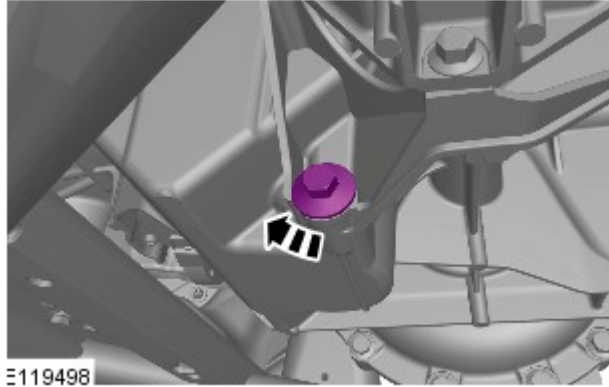


- *Torque: 10 Nm*

16.

 NOTE:

Install a new sealing washer.



Torque: 24 Nm

17.

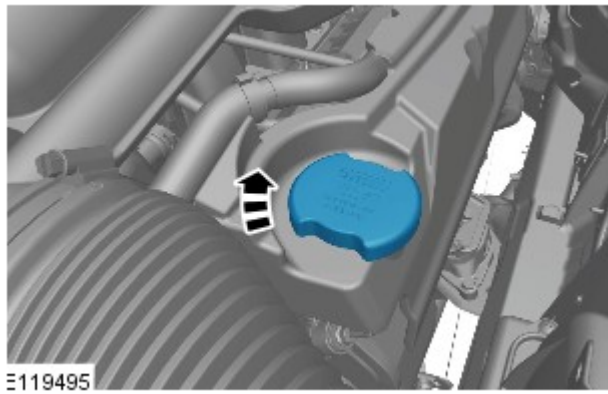
Connect the battery ground cable.

Refer to: Specifications (414-00 Battery and Charging System - General Information, Specifications).

18.

- Fill the engine with oil - for filling values on vehicles without supercharger:
Refer to: Specifications (303-01 Engine - V8 N/A 5.0L Petrol, Specifications).
- Fill the engine with oil - for filling values on vehicles with supercharger:
Refer to: Specifications (303-01 Engine - V8 S/C 5.0L Petrol, Specifications).
- Clean any residual engine oil from the oil filler cap area.

19.



20.

⚠ CAUTION:

Make sure that the vehicle has been left for 5 minutes from filling with oil.

- Start the engine and allow to run for 10 minutes, stop the engine.
- Check for leaks.

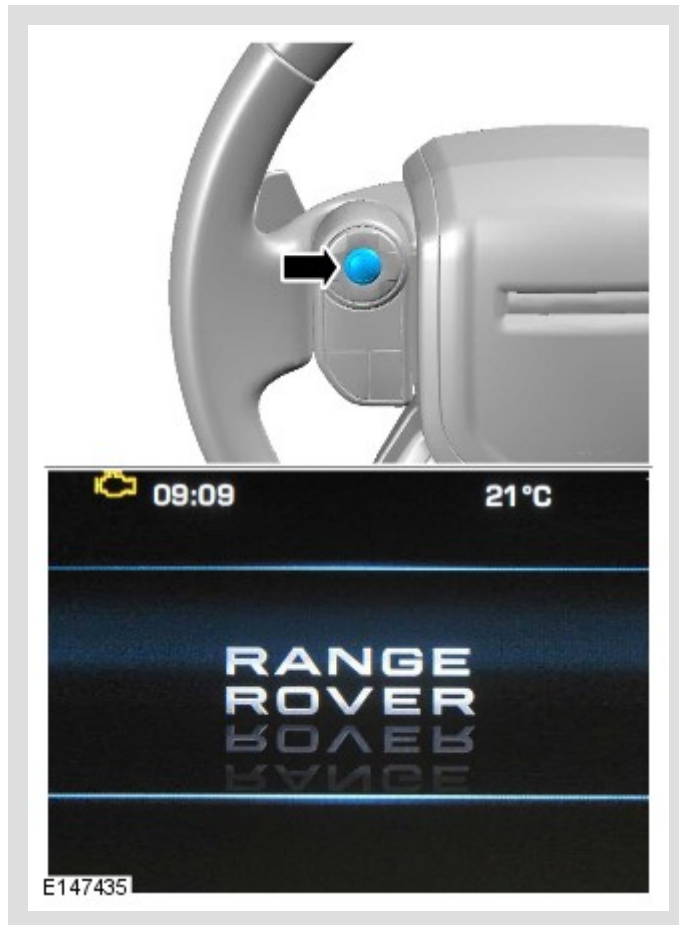
21.

⚠ CAUTIONS:

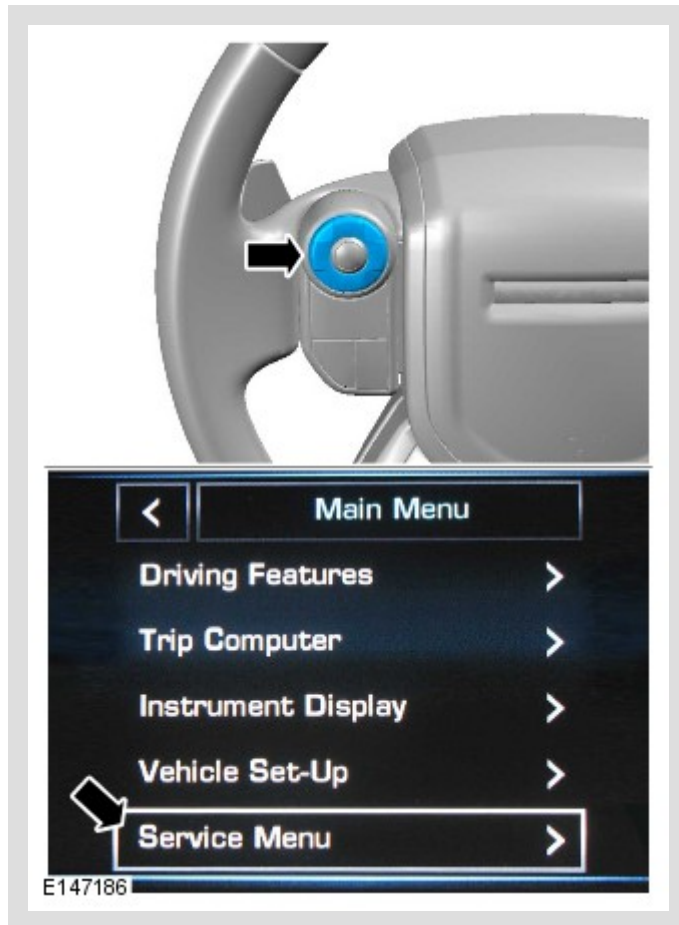
- Make sure that the selector lever is in the park (P) position.
- Make sure that the hood is open.

- Set the ignition to the ON position.

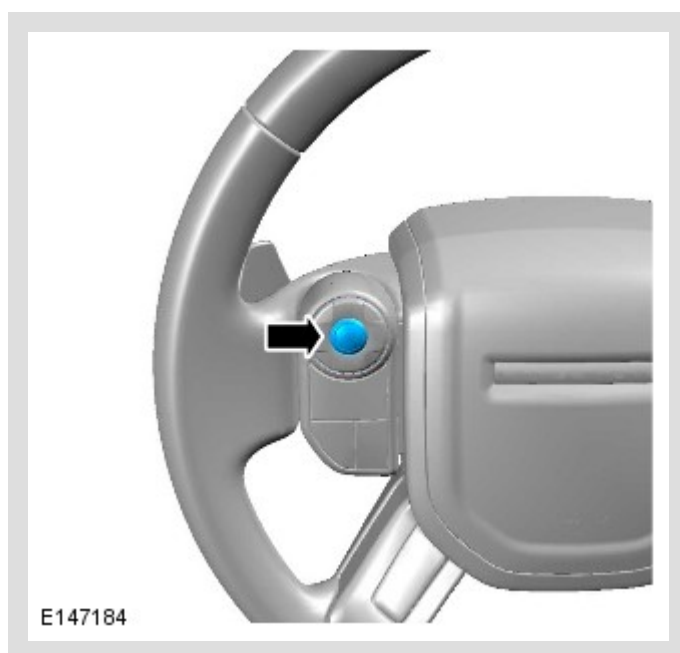
22.



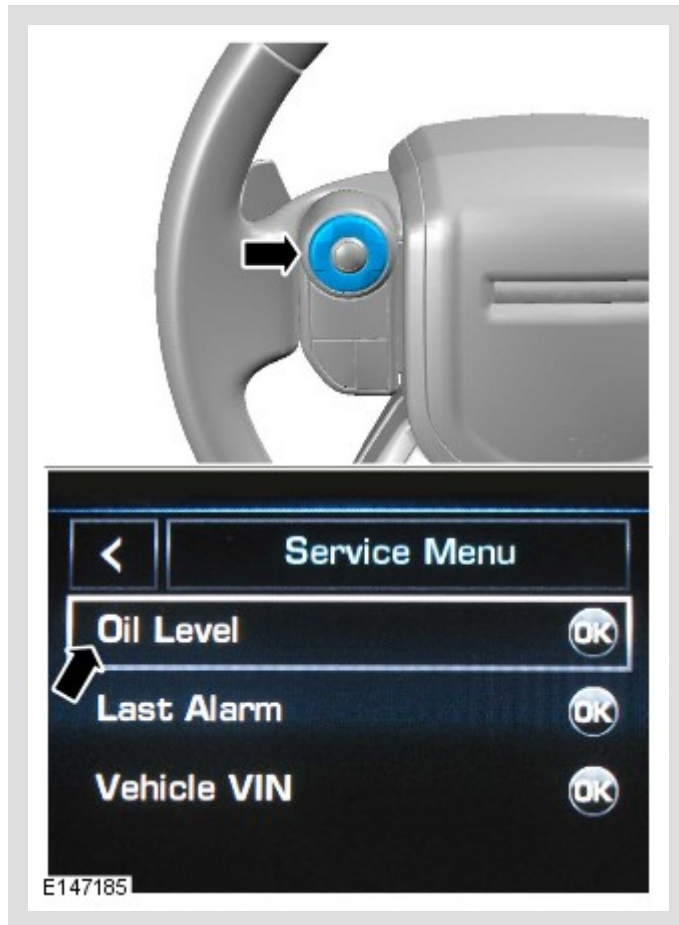
- Press the left-hand OK button to access the Instrument Cluster (IC) Main Menu.



- Press the left-hand directional button and select Service Menu.



- Press the left-hand OK button.



- Press the left-hand directional button and select the Oil Level display.



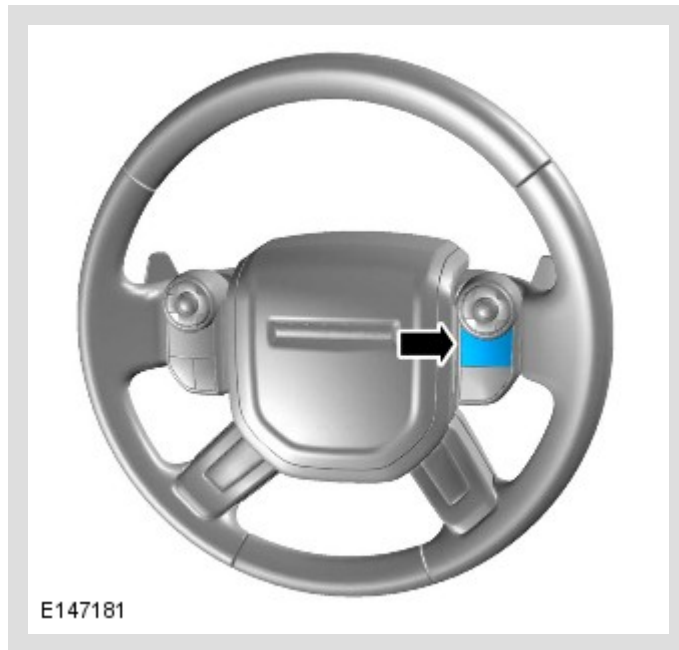
- Press the left-hand OK button to access the Oil Level display.

27.



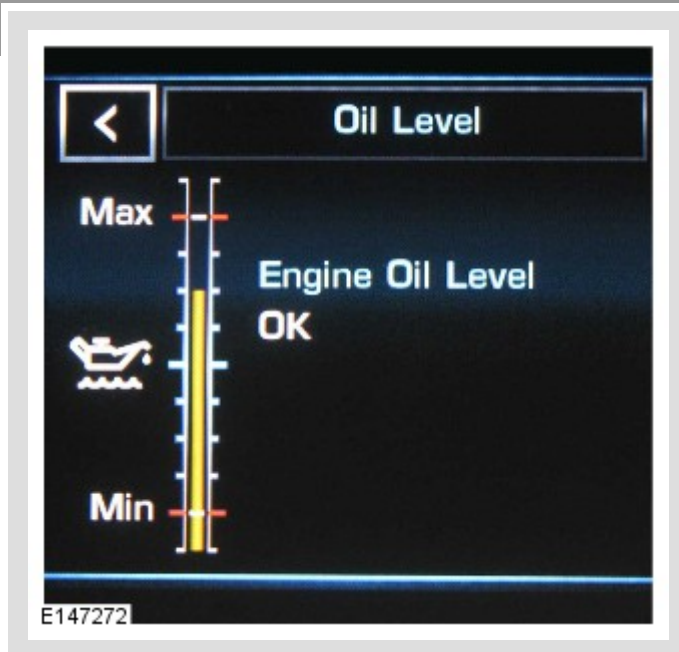
CAUTION:

Make sure that the hood is open.



- Press the speed control cancel button twice within 2 seconds while being on the oil level display.

28.



- The message center displays the actual oil level in service mode.
- Only after having started and run the engine for 15 minutes (as indicated in Step 20), switch off the engine, then stabilizing for 10 minutes, take a reading from the oil level display and, if necessary top up with engine oil.

29.

Set the ignition to the OFF position.

30. Allow 10 minutes for the engine oil level to stabilize if there has been additional oil top up.

 NOTE:

The following steps are to update the average oil level value.

31. Repeat step 21 to 27 to access the oil level display in service mode.

32.

 NOTE:

The actual volume displayed on the bar graph will be written into the electronic control unit as a new value.

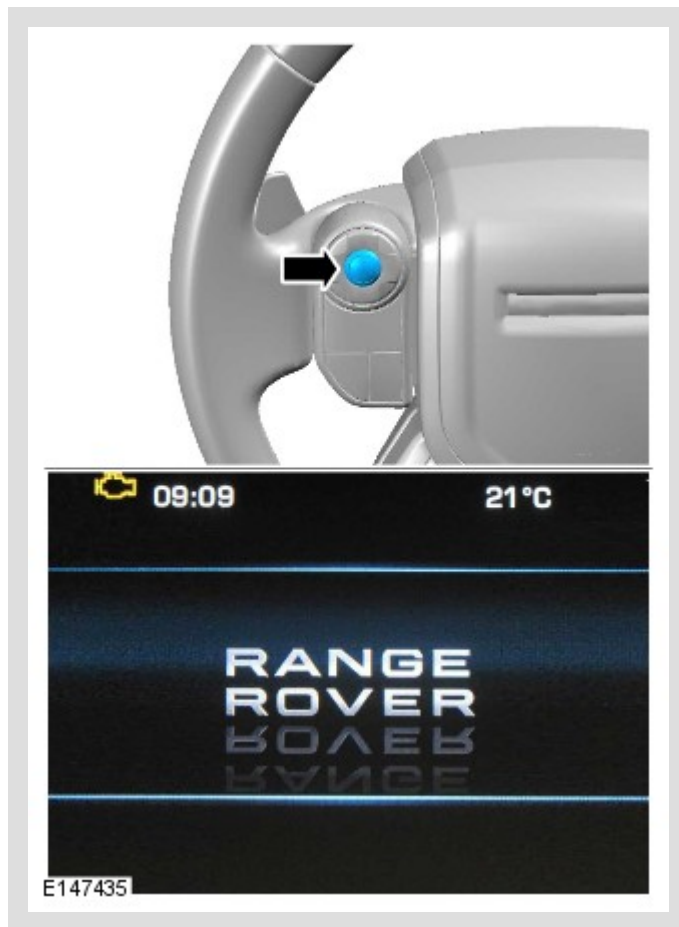


- Press and hold the speed control cancel button for more than 5 seconds.
- The message center display the current oil level.

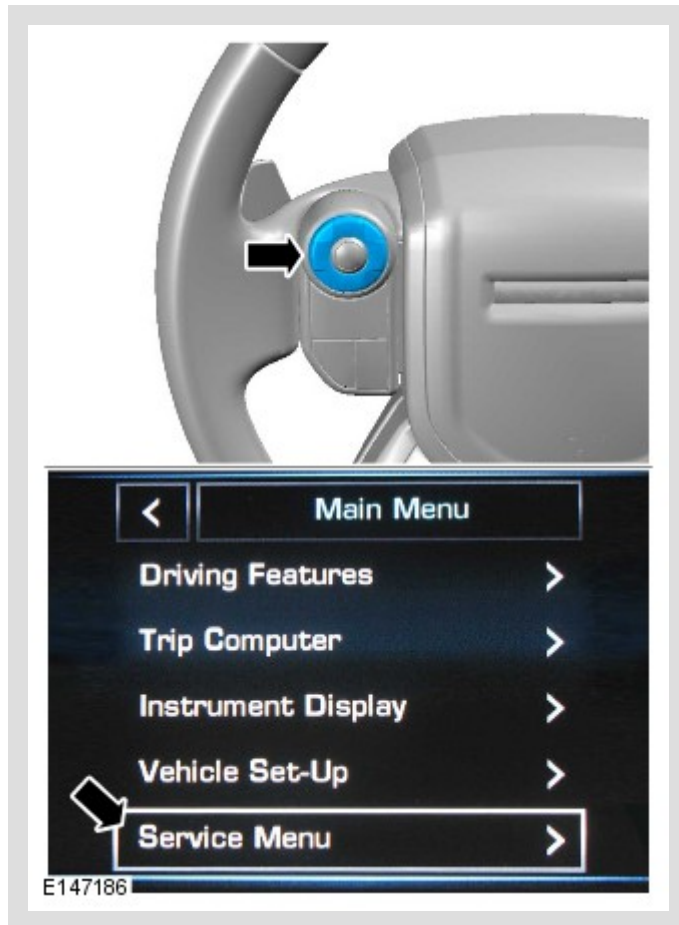
33. Set the ignition to the OFF position.

34. Set the ignition to the ON position.

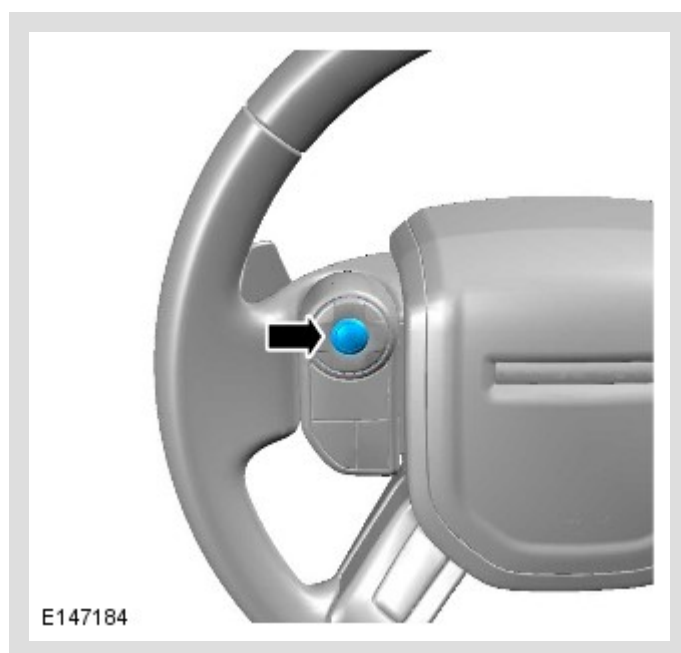
35.



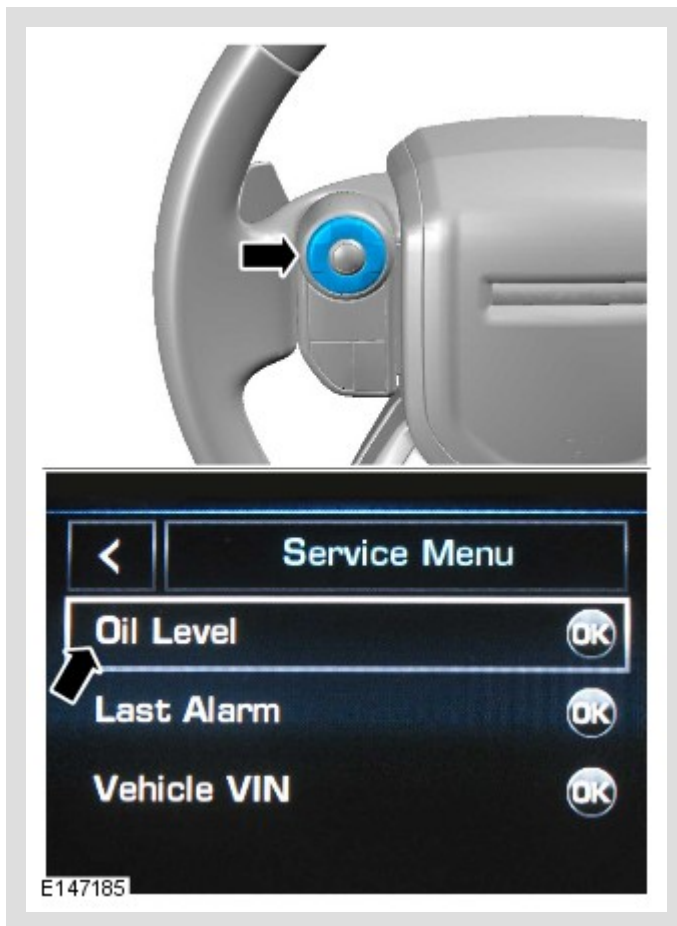
- Press the left-hand OK button to access the Instrument Cluster (IC) Main Menu.



- Press the left-hand directional button and select Service Menu.



- Press the left-hand OK button.



- Press the left-hand directional button and select the Oil Level display.



- Press the left-hand OK button and follow the instructions.

 NOTE:

If the message shows "Not available" wait for further 10 mins to allow the oil level to stabilize.

Make sure that the average oil level value has now been updated.

